

# NetPulse

Network Capacity Analytics & Compliance Reporting Platform

Prepared by [Your Name]  
[Team / Department]  
[Date]

**NetPulse converts raw interface-utilization exports into capacity intelligence and submission-ready reports.** It ingests routine polling files, computes per-link utilization, busy-period trends, growth forecasts and anomaly detection, and generates the threshold, upgrade-policy and capacity documentation that is currently assembled manually — with a complete, auditable evidence trail behind every figure. It is delivered as a single self-contained file requiring no server, no database, no licence and no network access.

**Zero**

INFRASTRUCTURE &  
LICENCE COST

**1M+**

ROWS PROCESSED PER  
DATASET

**54**

CONTROLLED RELEASES

**100%**

OFFLINE — NO DATA  
EGRESS

## BUSINESS OUTCOMES

- **Automated capacity reporting.** Threshold-breach, upgrade-policy and capacity reports that previously required approximately [X hours] of manual spreadsheet work each reporting cycle are now produced in seconds, complete with timestamped supporting evidence.
- **Submission-grade documentation.** Professional PDF reports containing reporting parameters, summary counts, per-day exceedance tables, a full evidence log and a prepared / reviewed / approved sign-off block — suitable for internal governance and regulatory submission.
- **Prevents unnecessary capital spend.** Links that appear saturated only because of stale bandwidth records are automatically flagged for verification instead of triggering an upgrade, avoiding procurement that is not required.
- **Prevents capacity-related incidents.** Genuinely congested links are identified early, with a statistical forecast of when each will exhaust its headroom and a recommended upgrade window.
- **No vendor dependency or spend.** The entire platform is one HTML file — no server, database, licence, subscription or third-party component.
- **Security and data residency by design.** Runs fully offline with all external network access blocked at the browser level; no operational data leaves the machine, avoiding the security-review and data-transfer overhead a hosted alternative would incur.
- **Reusable across the business.** Fully white-labelled and configurable — thresholds, market hours, organisation branding and user-defined fields — so the same platform serves other environments, teams and customers without code changes.

## DATA-INTEGRITY ISSUES IDENTIFIED AND RESOLVED

During validation against live data the platform surfaced several material defects in the source exports. Each is now detected automatically and surfaced to the operator, so capacity decisions rest on correct figures.

| ISSUE FOUND                                                                | BUSINESS CONSEQUENCE IF UNDETECTED                                               | RESOLUTION                                                                                                |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Stale bandwidth records</b> — links reporting above 100% utilization    | Healthy links appear saturated; unnecessary upgrade spend and false escalations  | Detected and flagged for capacity review; corrected capacity applied via override, all figures recomputed |
| <b>Inconsistent device naming</b> splitting one link across two identities | Traffic under-reported; a congested link can appear quiet and be missed entirely | Automatic and manual consolidation of naming variants                                                     |
| <b>Unit misinterpretation</b> in the vendor export headers                 | Throughput and utilization figures materially wrong in both directions           | Unit resolution corrected and validated against the source file                                           |
| <b>Timestamp inconsistency</b> between date formats in the same export     | Readings shifted in time; market-hours analysis and breach evidence unreliable   | Timestamps preserved exactly as recorded, with an explicit time-zone control                              |

| ISSUE FOUND                                   | BUSINESS CONSEQUENCE IF UNDETECTED        | RESOLUTION                                                                      |
|-----------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------|
| Threshold counts excluding off-hours activity | Overnight congestion invisible in reports | Market and non-market activity reported separately and together in one document |

**Net effect:** every published figure is now reconcilable line-by-line against the original source file, and the conditions that previously produced misleading capacity conclusions are detected automatically rather than relying on manual review.

## PLATFORM CAPABILITIES

### Data ingestion

- Multi-file Excel / CSV upload; columns matched by **meaning**, not fixed names, and fully editable
- Automatic unit normalisation across bps / Kbps / Mbps / Gbps / bytes
- Scales to 6-month, 1M+ row datasets via streaming ingestion
- Append-only history; duplicate records rejected, originals never overwritten
- Validation console with parse inspector before any data is committed

### Analytics

- Per-sample utilization measured against the capacity in force at that moment
- Busy-period (p95), average and peak analysis with configurable basis
- Statistical growth forecasting with confidence range and time-to-exhaustion
- Anomaly detection against each link's own behavioural baseline, with explanations
- Risk scoring weighted by business impact; market vs non-market hour comparison

### Reporting

- **Threshold Breach PDF** — six-month warning and critical counts with full timestamped evidence and sign-off block
- **Upgrade Policy workbook** — summary plus per-day and per-timestamp evidence sheets
- **Regulatory Capacity Report** — utilization against a configurable threshold with headroom, augmentation trigger and projected breach date
- **Processed-data export** — every computed value exposed for independent audit
- Executive, capacity, trend and business-impact reports

### Governance & configuration

- Append-only audit log of every upload, configuration change and export
- Configurable thresholds, market hours, upgrade policy and retention basis
- **User-defined custom fields** (e.g. ISP, Purpose) flowing through import, exports and reports
- Organisation branding and named prepared / reviewed / approved signatories
- Maintenance windows excluded from analytics and reporting

## ARCHITECTURE, SECURITY AND DELIVERY

### Architecture

A single self-contained HTML file. All processing occurs in the browser; data is held in local browser storage. There is no server, database, installation or administrative dependency — the file is opened directly and works immediately, including on an air-gapped machine.

No server
No database
No licence cost
No installation
  
Air-gap capable

### Security

External network access is blocked at the browser level by an enforced content-security policy. The platform makes no outbound calls of any kind — no telemetry, no external fonts, scripts or services. Operational data never leaves the machine on which it is opened.

### Quality assurance

Delivered across **54 controlled releases**, each validated by an automated regression suite exercising every view, export and interaction and asserting a zero-error run. Supported by a documented operations SOP and a full change history.

## PROPOSED NEXT STEPS

- Walkthrough and demonstration session with [Manager / Team]

- Extend coverage from core links to [additional links / sites / environments]
- Confirm reporting thresholds and document format with the compliance function
- Agree a standard reporting cadence and distribution list
- Evaluate reuse for [other customers / business units]

*Note: NetPulse provides the underlying data, calculations and document structure. Regulatory thresholds and report formats remain subject to validation by the compliance function against the currently applicable circulars.*